Ajay Venkatesh

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Education

New York University, New York, NY — *Aug 2024 – May 2026*

Master of Science in Computer Engineering (GPA: 3.9/4.0)

Coursework: System Design, Machine Learning, Deep Learning, Big Data, Advanced Java, Data Structures

Experience

- **Campus Mesh** — *Jan 2026 – Present*
  Software Engineer — New York, US
  - Developed scalable **Node.js microservices** powering real-time academic workflows for **10K+ users** across study groups, messaging, notifications, and profile management, backed by a **MongoDB** store.
  - Built a personalized **recommendation engine** using collaborative filtering and behavioral analytics, increasing engagement by **28%** and retention by **50%**.
  - Worked on **LLM-based retrieval pipelines (RAG)** for CampusIQ (AI Assistant), leveraging **embedding-based semantic search** and structured data indexing to improve contextual accuracy by **30%**.
- **Amazon** — *May 2025 – Aug 2025*
  Software Development Engineer Intern — Seattle, US
  - Deployed a **Retrieval-Augmented Generation (RAG)** pipeline with an **agentic workflow** that autonomously retrieves operational logs to reason over incident details, generating context-aware summaries and remediation steps.
  - Implemented a **Model Context Protocol (MCP)** server for efficient tool-augmented retrieval, cutting AWS Athena query time and compute costs by **40%**.
  - Built a high-throughput, event-driven incident triage system on **AWS ECS Fargate, SQS, SNS**, decoupling distributed message processing and reducing response latency by **50%**.
  - Provisioned resilient **Infrastructure-as-Code** using **AWS CDK** on auto-scaling ECS clusters, ensuring **99.9%** availability under 2x peak load.
- **New York University (Novel AI Technologies, Inc.)** — *Aug 2024 – Present*
  Software Developer — New York, US
  - Architected an **NSF-funded ($100K)** AI-powered health app for **iOS** and **Android** processing real-time wearable data; deployed cloud-hosted AI models for health metric analysis and food image processing.
  - Engineered **ETL pipelines** transforming **NoSQL** into **PostgreSQL** with **concurrency control and locking**; processed **250K+** records and powered **AWS QuickSight** dashboards.
  - Developed an insurance analytics platform (**React, Node.js, REST APIs**) deployed via **Docker** on **EC2**; implemented Stripe billing, RBAC, and row-level security, generating **$50K+** revenue across **6 clients**.
  - Streamlined underwriting via a broker portal integrated with insurer platforms; implemented **document/media processing** and **WebSocket-based real-time communication**.
- **PricewaterhouseCoopers (PwC)** — *Aug 2021 – Aug 2024*
  Senior Software Engineer — Bengaluru, India
  - Developed a distributed backend on **Microsoft Azure** for a health platform integrating with **Microsoft CRM**, with event-driven processing and fault-tolerant retries over **SQL Server** stores.
  - Deployed an **NLP-based OCR neural model** on Azure to automate structured data extraction, reducing processing time by **75%**.
  - Optimized data ingestion with **parallel processing** on **100K-row** Excel feeds, cutting execution from **5 hours to 1 hour**.
  - Led code reviews and mentored engineers on **software design principles**, reducing production bugs by **25%**.

Projects

- **Advanced Expense Tracker (Python Desktop Application)** | *Python, Tkinter, SQLite3, Pandas, Matplotlib*
  - Architected a modular **Python application** for individual and group expenses with secure login, applying **OOP principles** and **MVC & DAO design patterns** over **SQLite3**.
  - Implemented concurrent transaction handling and real-time cost splitting using **Python threading** and locks for thread-safe DB writes; visualized spending trends with **Matplotlib**.
- **Hospital Management System (Full-Stack Application)** | *React, Node.js, Express, MongoDB, SageMaker, Hugging Face*
  - Built a **full-stack healthcare platform** (React, Node.js, Express, MongoDB) managing appointments, patient records, lab reports, and medical history with **role-based access control**.
  - Fine-tuned a **Hugging Face transformer-based model** reaching **88% accuracy** for health condition prediction and deployed on **AWS SageMaker** for real-time diagnostic support.
- **InviLite** | *AWS, SageMaker, Bedrock, FAISS, Python*
  - Architected a cloud-native data pipeline (Kinesis, Lambda, S3, Glue, DynamoDB) ingesting structured and unstructured enterprise data into governed ML pipelines.
  - Built a **RAG system** (SVM + FAISS + **Bedrock LLM**), cutting incident resolution from **1–2 hours to minutes** and lifting root-cause classification accuracy from **56% to 92%**.

Skills

- **Programming Languages:** C++, Java, Python, TypeScript, JavaScript (ES6), SQL
- **Web & Backend:** React, Node.js, Express.js, REST API, GraphQL API, Microservices, HTML, CSS, Sequelize
- **Databases:** MySQL, PostgreSQL, DynamoDB, MongoDB, FireBase, Dataverse
- **Cloud & DevOps:** AWS (CDK, EC2, S3, DynamoDB, API Gateway, Lambda, Cognito, IAM, SageMaker), Azure (OpenAI, Function Apps, Storage), Docker, Kubernetes, CI/CD
- **AI/ML:** LLM Fine-tuning, RAG (LangChain, FAISS), NLP, Deep Learning, Reinforcement Learning, Hugging Face
- **Tools & Platforms:** Git, Linux, Visual Studio, Cursor, GitHub Copilot, Jupyter, Postman, Power BI

Publications

Supervised ML System Based Segmentation and Classification of Stroke Using Deep Learning Techniques *(IEEE)*

An Aquila-optimized SVM Classifier for Diabetes Prediction *(IEEE)*

Certifications

AWS Cloud Practitioner | Stanford Machine Learning (Coursera) | Azure Data Fundamentals